

SANDERSON FIELD, WA, USA

WMO: 727925

Lat: 47.238N

Lon: 123.141W

Elev: 271

StdP: 14.55

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 94227

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.1	25.8	13.3	10.9	27.5	19.0	14.5	30.8	24.5	47.9	20.9	47.3	0.9	40	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.2	88.3	65.7	83.9	64.4	79.9	63.0	67.4	84.8	65.7	81.7	63.9	77.6	7.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	81.3	69.9	59.5	76.7	68.9	58.3	73.3	67.6	31.8	84.7	30.5	81.7	29.2	77.5	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Max
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(p)
(4) 20.5	18.3	16.4		15.7	96.0	5.4	3.8	11.9	98.7	8.7	101.0	5.7	103.1	1.8	105.9	
(5)			DB	14.8	71.2	5.2	3.3	11.0	73.6	7.9	75.5	5.0	77.3	1.2	79.7	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	39.7	40.6	43.8	47.9	54.3	59.2	64.3	64.8	59.5	50.6	43.4	39.0	(6)
	DBStd	10.57	5.81	5.04	4.74	5.14	5.51	5.24	4.68	4.29	4.64	4.93	6.20	5.81	(7)
	HDD50	1508	320	264	198	99	20	1	0	0	1	52	210	343	(8)
	HDD65	5356	783	683	656	513	335	189	69	57	173	445	647	806	(9)
	CDD50	1749	2	1	7	36	152	276	445	458	286	72	13	1	(10)
	CDD65	124	0	0	0	0	2	15	48	50	9	0	0	0	(11)
	CDH74	2451	0	0	2	26	134	330	830	827	298	3	1	0	(12)
	CDH80	867	0	0	0	4	32	113	317	313	88	0	0	0	(13)
(14) Wind	WSAvg	5.9	4.4	4.9	5.9	6.4	7.2	7.7	7.7	7.1	5.4	4.6	4.5	4.3	(14)
(15) Precipitation	PrecAvg	64.2	10.4	6.9	7.2	4.5	2.8	1.9	0.7	1.4	2.3	6.5	10.6	10.3	(15)
	PrecMax	85.6	21.8	19.5	17.4	9.9	5.6	3.5	2.7	4.1	9.0	16.1	25.4	18.3	(16)
	PrecMin	48.2	4.7	0.0	1.4	1.5	0.2	0.3	0.1	0.0	0.0	0.2	3.7	3.2	(17)
	PrecStd	11.3	3.7	3.6	3.3	2.1	1.6	0.9	0.7	1.2	2.2	3.8	4.8	4.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	58.5	68.3	77.7	84.4	90.2	95.3	93.5	88.1	72.9	60.9	56.3	(19)
		MCWB	52.4	47.0	53.4	57.7	61.9	66.8	68.2	67.1	63.8	57.4	56.9	53.5	(20)
	2%	DB	52.4	54.2	61.1	69.6	77.7	83.0	87.9	88.0	81.8	66.5	56.9	52.1	(21)
		MCWB	50.1	47.2	50.5	54.7	60.1	64.0	65.8	66.2	62.8	55.4	53.4	50.1	(22)
	5%	DB	50.3	51.6	56.1	63.4	71.6	76.9	82.6	83.0	76.6	62.7	54.7	49.6	(23)
		MCWB	48.0	46.7	48.2	51.2	57.8	61.5	64.9	65.0	61.3	55.3	51.9	47.4	(24)
	10%	DB	48.3	49.1	52.7	58.4	66.1	71.5	78.1	78.1	71.8	59.5	52.5	47.2	(25)
		MCWB	46.3	45.7	47.1	49.3	54.9	59.1	63.1	63.3	59.6	54.6	50.2	45.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.7	51.9	55.2	59.7	63.8	68.2	70.8	69.3	67.2	61.1	57.4	54.0	(27)
		MCDB	54.8	54.2	63.2	74.4	79.8	87.1	91.1	88.2	81.5	66.0	60.2	55.7	(28)
	2%	WB	50.8	49.8	52.0	55.4	60.8	64.9	67.3	67.3	64.3	58.7	54.4	50.4	(29)
		MCDB	51.8	51.9	58.1	67.6	74.9	80.7	84.5	84.9	76.8	62.7	56.3	51.8	(30)
	5%	WB	48.4	47.7	50.1	52.6	58.3	62.0	65.5	65.5	62.4	56.8	52.7	47.8	(31)
		MCDB	49.7	50.1	54.6	60.4	69.9	74.9	81.7	81.1	72.8	60.6	54.3	49.2	(32)
	10%	WB	46.4	46.1	48.1	50.6	56.1	59.6	63.5	63.8	60.7	55.1	50.6	45.9	(33)
		MCDB	47.9	48.5	52.1	56.7	64.9	70.1	77.1	77.0	69.5	58.6	52.1	47.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	14.6	16.6	19.1	20.4	20.6	23.8	24.2	23.1	17.7	13.5	10.5	(35)
		MCDBR	12.8	19.8	24.9	31.0	32.9	33.4	35.4	36.3	35.6	24.7	15.1	11.1	(36)
		MCWBR	10.2	13.4	15.2	16.8	17.0	15.9	16.0	16.3	18.0	15.5	11.2	9.3	(37)
	5% WB	MCDBR	10.4	14.0	19.7	24.5	28.0	29.2	32.3	31.4	28.3	18.3	11.7	10.2	(38)
		MCWBR	8.9	10.6	13.1	14.1	14.9	14.3	15.0	14.6	15.1	12.7	9.6	8.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.331	0.321	0.336	0.367	0.393	0.376	0.361	0.363	0.348	0.338	0.335	0.331	(40)
	taud		2.446	2.526	2.497	2.378	2.322	2.399	2.463	2.460	2.504	2.529	2.457	2.397	(41)
	Ebn at Noon		238	271	282	281	277	282	285	279	274	260	233	220	(42)
	Edh at Noon		22	24	29	36	39	36	34	32	28	24	21	20	(43)
(44) All-Sky Solar Radiation	RadAvg		323	594	886	1297	1643	1796	2006	1717	1260	712	379	266	(44)
	RadStd		49	81	107	123	137	185	154	120	117	79	50	39	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-1.44	-3.87	N/A	N/A	N/A	N/A	N/A	+257	+56
Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page